

Adam F. Holt

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Employment

University of Miami Assistant Professor Department of Marine Geosciences, Rosenstiel School of Marine, Atmospheric, and Earth Science	Aug. 2019 – present
Massachusetts Institute of Technology Postdoctoral Associate/Fellow Department of Earth, Atmospheric and Planetary Sciences	Aug. 2016 – Jul. 2019

Education

University of Southern California , Los Angeles Ph.D. in Geological Sciences Advisor: Thorsten W. Becker Dissertation title: Trench migration, slab bending, and mantle flow at subduction zones	2011 – 2016
Imperial College , London M.Sci. Geophysics (First-class honors)	2007 – 2011

Publications

Conrad, E. M., Faccenna, C., **Holt, A. F.**, and Becker, T. W.: 2024. Tectonic reorganization of the Caribbean plate system in the Paleogene driven by slab interactions and mantle convection. *Geochem., Geophys., Geosys.*, doi:10.1029/2024GC011499.

Epstein, G. S., Condit, C. B., Stoner, R. K., **Holt, A. F.**, and Guevara, V.: 2021. Evolving Subduction Zone Thermal Structure Drives Extensive Forearc Mantle Wedge Hydration. *AGU Advances*, 5, doi:10.1029/2023AV001121.

Goldberg, S. L., and **Holt, A. F.**: 2024. Characterizing the complexity of subduction zone flow with an ensemble of multiscale global convection models. *Geochem., Geophys., Geosys.*, 25, doi:10.1029/2023GC011134.

Turino, V., and **Holt, A. F.**: 2024. Spatio-temporal variability in slab temperature within dynamic 3-D subduction models. *Geophys. J. Int.*, doi:10.1093/gji/ggad489.

Clennett, E. J., **Holt, A. F.**, Tetley, M. G., Becker, T. W., Faccenna, C.: 2023. Assessing plate reconstruction models using plate driving force consistency tests. *Sci. Rep.*, doi:10.1038/srep2300487

Holt, A. F., The Topographic Signature of Mantle Pressure Build-Up Beneath Subducting Plates: Insights from Spherical Subduction Models. *Geophys. Res. Lett.*, doi:10.1029/2022gl100330.

- Behr, W. M., **Holt, A. F.**, Becker, T. W., Faccenna, C.: 2022. The effects of plate interface rheology on subduction kinematics and dynamics. *Geophys. J. Int.*, 230, doi:10.1093/gji/ggac075.
- Holt, A. F.**, and Condit, C. B.: 2021. Slab temperature evolution over the lifetime of a subduction zone. *Geochem., Geophys., Geosys.*, 22, doi:10.1029/2020GC009476.
- Faccenna, C., Becker, T. W., **Holt, A. F.**, and Brun, J-P.: 2021. Mountain building, mantle convection, and supercontinents: Holmes (1931) revisited. *Earth Planet. Sci. Lett. (Frontiers)*, 564, doi: 10.1016/j.epsl.2021.116905
- Royden, L. H., and **Holt, A. F.**: 2020. Subduction dynamics and mantle pressure: (i) An Analytical Framework Relating Subduction Geometry, Plate Motion, and Asthenospheric Pressure. *Geochemistry, Geophysics, Geosystems.*, doi: 10.1090/2020GC009032, 2020.
- Holt, A. F.**, and Royden, L. H.: 2020. Subduction dynamics and mantle pressure: (ii) Towards a Global Understanding of Slab Dip and Upper Mantle Circulation. *Geochemistry, Geophysics, Geosystems.*, doi: 10.1002/2019GC008771, 2020.
- Holt, A. F.**, Royden, L. H., Becker, T. W., Faccenna, C.: Slab interactions in 3-D subduction settings: The Philippine Sea Plate region. *Earth and Planetary Science Letters*. 489, 72-83, 2018.
- Király, A., **Holt, A. F.**, Funicello, C., Capitanio, F., Faccenna, C.: Modeling slab-slab interactions: Dynamics of a double-sided subduction system. *Geochemistry, Geophysics, Geosystems.*, doi: 10.1002/2017GC007199, 2018.
- Faccenna, C., **Holt, A. F.**, Becker, T. W., Lallemand, S., Royden, L. H.: Dynamics of the Ryukyu/Izu-Bonin-Marianas double subduction system. *Tectonophysics.*, doi:10.1016/j.tecto.2017.08.011, 2017.
- Holt, A. F.**, Royden, L., and Becker, T. W.: The dynamics of double slab subduction. *Geophys. J. Int.*, 209 250-265, 2017.
- Faccenna, C., Oncken, O., **Holt, A. F.**, and Becker, T. W.: Initiation of the Andean Orogeny by lower mantle subduction. *Earth Planet. Sci. Lett.*, 463, 189-201, 2017.
- Holt, A. F.** and Becker, T. W.: The effect of a power-law mantle viscosity on trench retreat rate. *Geophys. J. Int.*, doi:10.1093/gji/ggw392, 2016.
- Holt, A. F.**, Buffett, B. A., and Becker, T. W.: Overriding plate thickness control on subducting plate curvature. *Geophys. Res. Lett.*, 42, 3802-3810, doi:10.1002/2015GL063834, 2015.
- Jagoutz, O., Royden, L., **Holt, A. F.**, and Becker, T. W.: Anomalously fast convergence of India and Eurasia caused by double subduction. *Nature Geoscience*, 8, 475-478, doi: 10.1038/NGEO2418, 2015.
- Holt, A. F.**, Becker, T. W., and Buffett, B. A.: Subduction dynamics and overriding plate stress in thermo-mechanical subduction models. *Geophys. J. Int.*, 201, 172-192, doi: 10.1093/gji/ggv011, 2015.
- Sun, D., Miller, M. S., **Holt, A. F.**, and Becker, T. W.: Hot upwelling conduit beneath the Atlas Mountains, Morocco. *Geophys. Res. Lett.*, 41, 8037-8044, doi:10.1002/2014GL061884, 2014.

Teaching Experience

Course Instructor (University of Miami)

Natural Disasters: Hollywood vs. Reality (GSC 107)	Spring 2020, 2021, 2022, 2023; Fall 2024
Geodynamics (MGS 723)	Fall 2021, 2023
Summer Field Geology (GSC 580, co-instructor w/James Klaus)	Summer 2023
Subduction Zone Geodynamics (MGS 710)	Fall 2024

Grants and Awards (*since beginning UM faculty position, 2019*)

NSF-EAR Petrology and Geochemistry: Collaborative Research: Examining the paleo-thermal structure of the Alpine subduction interface with an integrated petrochronology and geodynamic modeling approach. Co-PI (UM: \$362,349)	2024
UM Provost's Early Career Award for Faculty Excellence	2023
AGU Editors' Citation for Excellence in Refereeing	2022
NSF-EAR Geophysics: Collaborative Research: Probing feedbacks between thermal structure, petrologic transformation, and rheologic evolution within dynamically evolving subduction zones. Lead PI (UM: \$287,776)	2021
NSF-EAR Geophysics: Constraining multi-scale interactions between slabs and mantle flow within Western Pacific subduction zones, 06/2022-06/2025. Sole PI (UM: \$349,874).	2021
XSEDE/ACCESS Research Allocations (supercomputing resources) Recent allocation (2023): \$21,474 worth of compute time/storage.	yearly 2018-2024

Service and Mentorship

Internal (University of Miami):

Ph.D. supervision: Valeria Turino (2020-present), Tao Zhao (from Fall '23), Zonglin Guo (From Fall '23)

Postdoc supervision: Samuel Goldberg (NSF EAR Postdoc fellowship; 2021-2022), Ryan Stoner (2022-present), Gabe Epstein (NSF EAR Postdoc fellowship; from 2023-present).

Ph.D. committee: Farzaneh Zanjani, Bhuvan Varugu, Sara Mirzaee.

Undergraduate mentees: Chantal Newallo (2020), Jazmin Garza (2021-2022), Reid Jansen (2022-present).

External:

External Ph.D. examiner: Roma Tre University [Italy]; Monash University [Australia]; Australia National University [Australia].

Proposal reviews: NSF EAR Earthscope/GeoPRISMS/CSEDI/Geophysics/Marine Geology and Geophysics/Postdoctoral Fellowships, DFG (German Research Foundation), RSF (Russian Science Foundation).

Manuscript reviews: Nature; Nature Geosci.; Nature Comm.; Geology; J. Geophys. Res.; Geochem., Geophys., Geosyst.; Tectonics; Phys. Earth Planet. Int.; Geophys. J. Int.; Scientific Reports; Tektonika; Geophys. Res. Lett.; Earth Planet. Sci. Lett.; Progress in Earth and Planet. Sci.; Tectonophysics; Solid Earth; Frontiers; Science Advances.

Workshops: Session convenor and workshop co-organizer - AGU/SEG convergent margins workshop (2022); Session convenor - GSA annual meeting (2022).

Committees: Computational Infrastructure for Geodynamics (CIG) Science Steering Committee and Education Working Group (2023-).